

nick mckenna

Email: ↪ nmckenna363@gmail.com
Homepage: ↪ www.nmckenna.me

Summary

- Applied Researcher at GitHub Copilot.
Ex-Research Scientist at Microsoft Research.
- Publications in NLP/AI venues, including the *Best Paper Award* at AACL 2023.
- Research Experience:
LLMs for code generation, planning, & agentic systems. Integrating LLMs with external data and systems. Language understanding tasks, hallucination, and knowledge bases.

Education

The University of Edinburgh

Ph.D. Informatics 2023
Thesis: *Inference of Natural Language Predicates in the Open Domain*
Advisor: Mark Steedman

M.Sc. Artificial Intelligence, *Distinction* 2019

Brown University

B.Sc. Computer Science 2017

Professional Experience

Applied Researcher III, GitHub Copilot London, UK, 2025.04 → Present

- Improving GitHub Copilot editor completions with novel contextualization and RAG techniques.

Research Scientist, Microsoft Research Cambridge, UK, 2024.02 → 2025.04

- Researched LLM code generation for multi-step planning and calculation in Microsoft Copilot use-cases. Affiliated with Calc Intelligence (Excel AI Research) and ENCODE teams.
- Lead project to develop novel training techniques for low-resource programming languages (e.g. Excel Formulas) and APIs, useful for integrating code generating LLMs with native applications.
- Developed synthetic data generation techniques which were adopted by sister team.

Applied Scientist II Intern, Amazon, Alexa AI Cambridge, UK, 2022.06 → 2022.11

- Developed a prototype for Alexa’s trivia question-answering, and published the research findings.
- Improved neural model architecture which integrates an LLM with a Differentiable Knowledge Graph, mitigating factual hallucination in responses.

Product Manager, TapToBook (Startup) Miami, USA, 2017.06 → 2018.08

- Identified business needs, researched and designed solutions, and prioritized the product roadmap for the 10-person development team (spanning web, iOS, and Android).
- Scaled the platform to bring in national brands including Planet Fitness, leading to profitability.

Skills

- **Modeling:** Program synthesis using code-generating LLMs for calculations and dynamic UX; Generative modeling of language tasks using LLMs and RAG over KBs and free text; Generation of synthetic data for targeted model finetuning; Bespoke neural architecture design and classical ML.
- **Languages:** Python, Java, Swift/iOS, C, SQL+Splunk
- **Tools:** PyTorch, Huggingface Transformers, LLMs like LLaMA & GPT-4, AWS, Git, Numpy, Scipy

Awards & Honors

The Best Paper Award at AACL 2023: <i>Smoothing Entailment Graphs with Language Models</i>	2023
Huawei Ph.D. Scholarship Award	2019
Outstanding M.Sc. Dissertation: <i>Learning Negation Scope Semantics with Structure</i> ↪ Link	2019

Scientific Service

Conference Reviewer

ICLR	2024
ARR (ACL)	2024 – Present
*ACL Conferences	2020 – 2023
COLING	2020

Talk Panelist

“Starting in NLP Research”	NAACL, 2021
----------------------------	-------------

Teaching

Graduate TA at the University of Edinburgh

Accelerated NLP	2019 – 2021
NLU and Machine Translation	2020 – 2022

Undergraduate TA at Brown University

Computational Linguistics	2017
Computer Graphics	2016
Computer Architecture	2015

Research Publications

TBA 2025	Nick McKenna , Xinnuo Xu, Jack Williams, Nicholas Wilson, Benjamin Van Durme, Christian Poelitz. <i>Synthetic Function Demonstrations Improve Generation in Low-Resource Programming Languages</i> . ↪ Link
TBA 2025	Christian Poelitz, Nick McKenna . <i>Synthetic Clarification and Correction Dialogues about Data-Centric Tasks: A Teacher-Student Approach</i> . ↪ Link
TBA 2025	Jinyang Li, Jack Williams, Nick McKenna , Arian Askari, Nicholas Wilson, Reynold Cheng. <i>Exploring Training-Free Knowledge Distillation for SLMs in Data Science Code Generation</i> .
AAAI 2025	Bhuvanashree Murugadoss, Christian Poelitz, Ian Drosos, Vu Le, Nick McKenna , Carina Suzana Negreanu, Chris Parnin, and Advait Sarkar. <i>Evaluating the Evaluator: Measuring LLMs’ Adherence to Task Evaluation Instructions</i> . ↪ Link
EMNLP Findings 2023	Nick McKenna* and Tianyi Li*; Liang Cheng, Mohammad Javad Hosseini, Mark Johnson, and Mark Steedman. <i>Sources of Hallucination by Large Language Models on Inference Tasks</i> . ↪ Link
AACL 2023	Nick McKenna , Tianyi Li, Mark Johnson, and Mark Steedman. <i>Smoothing Entailment Graphs with Language Models</i> . * <u>Best Paper Award</u> * ↪ Link
SustainNLP 2023	Nick McKenna and Priyanka Sen. <i>KGQA Without Retraining</i> . Workshop at ACL. ↪ Link
EMNLP 2021	Nick McKenna , Liane Guillou, Mohammad Javad Hosseini, Sander Bijl de Vroe, Mark Johnson, and Mark Steedman. <i>Multivalent Entailment Graphs for Question Answering</i> . ↪ Link
CASE 2021	Sander Bijl de Vroe* and Liane Guillou*; Miloš Stanojević, Nick McKenna , and Mark Steedman. <i>Modality and Negation in Event Extraction</i> . Workshop at ACL. ↪ Link
*SEM 2020	Nick McKenna and Mark Steedman. <i>Learning Negation Scope from Syntactic Structure</i> . ↪ Link